

2 Variable Maxima Minima

Two variable Maxima Minima (13.8)

When you have a function, $f(x, y) =$

Find, f_x which will be equal to zero.

f_y " " " " " "

$$\therefore f_x = 0 ; f_y = 0$$

Now, simultaneously solve the 2 equation to find the value of x and y which will be critical point.

Find f_{xx} and put the value of critical point, $f_{xx}(CP)$

" f_{xy} " " " " " " , $f_{xy}(CP)$

" f_{yy} " " " " " " , $f_{yy}(CP)$

$$D = f_{xx}(CP) \times f_{yy}(CP) - \{f_{xy}(CP)\}^2$$

Theory

13.8.6 THEOREM (*The Second Partial Test*) Let f be a function of two variables with continuous second-order partial derivatives in some disk centered at a critical point (x_0, y_0) , and let

$$D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$$

- (a) If $D > 0$ and $f_{xx}(x_0, y_0) > 0$, then f has a relative minimum at (x_0, y_0) .
- (b) If $D > 0$ and $f_{xx}(x_0, y_0) < 0$, then f has a relative maximum at (x_0, y_0) .
- (c) If $D < 0$, then f has a saddle point at (x_0, y_0) .
- (d) If $D = 0$, then no conclusion can be drawn.

Saddle Point

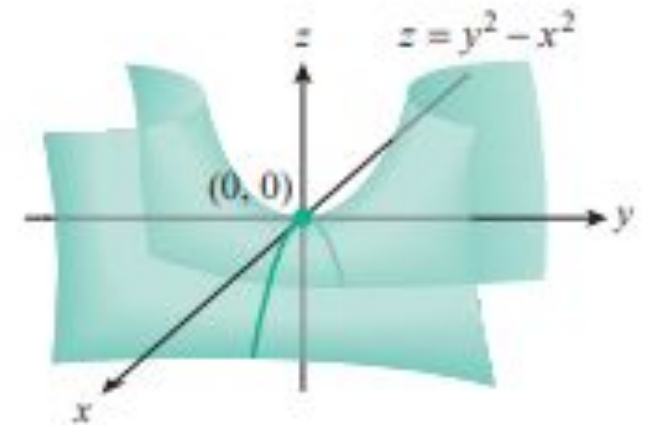
This function, whose graph is the hyperbolic paraboloid shown in Figure 13.8.5, has a critical point at $(0, 0)$, since

$$f_x(x, y) = -2x \quad \text{and} \quad f_y(x, y) = 2y$$

from which it follows that

$$f_x(0, 0) = 0 \quad \text{and} \quad f_y(0, 0) = 0$$

However, the function f has neither a relative maximum nor a relative minimum at $(0, 0)$. For obvious reasons, the point $(0, 0)$ is called a *saddle point* of f . In general, we will say that a surface $z = f(x, y)$ has a *saddle point* at (x_0, y_0) if there are two distinct vertical planes through this point such that the trace of the surface in one of the planes has a relative maximum at (x_0, y_0) and the trace in the other has a relative minimum at (x_0, y_0) .



The function $f(x, y) = y^2 - x^2$ has neither a relative maximum nor a relative minimum at the critical point $(0, 0)$.

▲ Figure 13.8.5

Practice Problems

► **Example 3** Locate all relative extrema and saddle points of

$$f(x, y) = 3x^2 - 2xy + y^2 - 8y$$

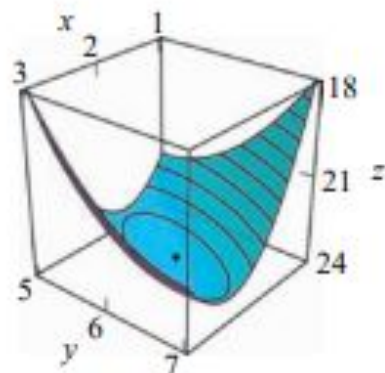
Solution. Since $f_x(x, y) = 6x - 2y$ and $f_y(x, y) = -2x + 2y - 8$, the critical points of f satisfy the equations

$$6x - 2y = 0$$

$$-2x + 2y - 8 = 0$$

Solving these for x and y yields $x = 2$, $y = 6$ (verify), so $(2, 6)$ is the only critical point. To apply Theorem 13.8.6 we need the second-order partial derivatives

$$f_{xx}(x, y) = 6, \quad f_{yy}(x, y) = 2, \quad f_{xy}(x, y) = -2$$



$$f(x, y) = 3x^2 - 2xy + y^2 - 8y$$

At the point $(2, 6)$ we have

$$D = f_{xx}(2, 6)f_{yy}(2, 6) - f_{xy}^2(2, 6) = (6)(2) - (-2)^2 = 8 > 0$$

and

$$f_{xx}(2, 6) = 6 > 0$$

so f has a relative minimum at $(2, 6)$ by part (a) of the second partials test. Figure 13.8.7 shows a graph of f in the vicinity of the relative minimum. ◀

► **Example 4** Locate all relative extrema and saddle points of

$$f(x, y) = 4xy - x^4 - y^4$$

Solution. Since

$$\begin{aligned} f_x(x, y) &= 4y - 4x^3 \\ f_y(x, y) &= 4x - 4y^3 \end{aligned} \quad (1)$$

the critical points of f have coordinates satisfying the equations

$$\begin{aligned} 4y - 4x^3 &= 0 & \text{or} & & y &= x^3 \\ 4x - 4y^3 &= 0 & & & x &= y^3 \end{aligned} \quad (2)$$

Substituting the top equation in the bottom yields $x = (x^3)^3$ or, equivalently, $x^9 - x = 0$ or $x(x^8 - 1) = 0$, which has solutions $x = 0$, $x = 1$, $x = -1$. Substituting these values in the top equation of (2), we obtain the corresponding y -values $y = 0$, $y = 1$, $y = -1$. Thus, the critical points of f are $(0, 0)$, $(1, 1)$, and $(-1, -1)$.

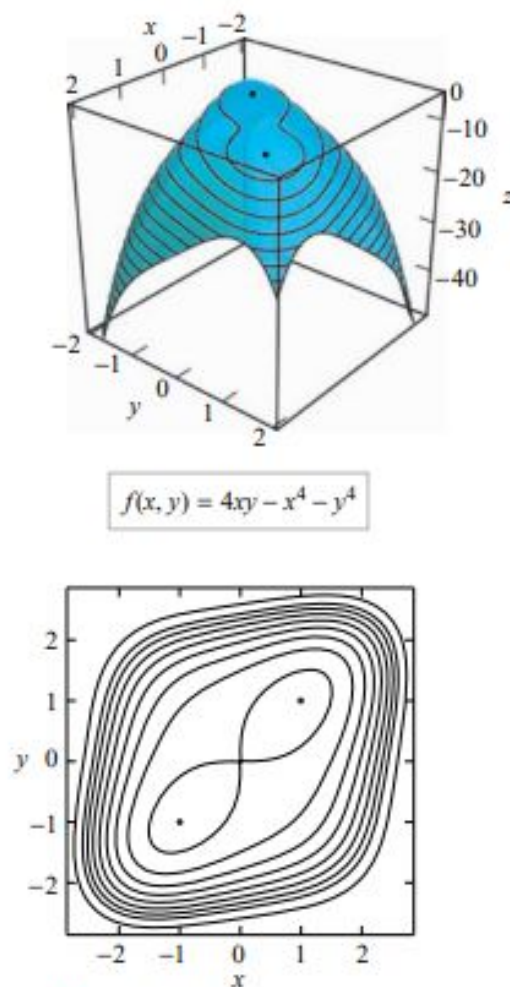
From (1),

$$f_{xx}(x, y) = -12x^2, \quad f_{yy}(x, y) = -12y^2, \quad f_{xy}(x, y) = 4$$

which yields the following table:

CRITICAL POINT (x_0, y_0)	$f_{xx}(x_0, y_0)$	$f_{yy}(x_0, y_0)$	$f_{xy}(x_0, y_0)$	$D = f_{xx}f_{yy} - f_{xy}^2$
$(0, 0)$	0	0	4	-16
$(1, 1)$	-12	-12	4	128
$(-1, -1)$	-12	-12	4	128

At the points $(1, 1)$ and $(-1, -1)$, we have $D > 0$ and $f_{xx} < 0$, so relative maxima occur at these critical points. At $(0, 0)$ there is a saddle point since $D < 0$. The surface and a contour plot are shown in Figure 13.8.8. ◀



▲ Figure 13.8.8

9. $f(x, y) = y^2 + xy + 3y + 2x + 3$

$$f_x = 0 + y + 0 + 2 + 0$$
$$= y + 2$$

$$f_y = 2y + x + 3 + 0 + 0$$
$$= 2y + x + 3$$

$$\therefore f_x = 0$$

$$\therefore y + 2 = 0$$

$$\therefore y = -2 \quad \text{--- (1)}$$

$$f_y = 0$$

$$2y + x + 3 = 0$$

putting the value
of eqⁿ (1)

$$\therefore 2(-2) + x + 3 = 0$$

$$\Rightarrow -4 + x + 3 = 0$$

$$\Rightarrow x = 1$$

$$y = -2, \quad x = 1$$

$\therefore$ critical point $(1, -2)$

$$f_{xx} = 0$$

$$f_{xy} = 1$$

$$f_{yy} = 2$$

$$D = f_{xx}(CP) f_{yy}(CP) - \{f_{xy}(CP)\}^2$$
$$= 0 \times 2 - (1)^2$$
$$= -1$$

Since $D < 0$ and ~~$f_{xx}(CP) = 0$~~ ,
then there is a saddle point at
 $(1, -2)$

12. $f(x, y) = xy - x^3 - y^2$

$$f_x = y - 3x^2 - 0 = y - 3x^2$$

$$f_y = x - 0 - 2y = x - 2y$$

$$f_x = 0$$

$$\therefore y - 3x^2 = 0$$

$$\therefore y = 3x^2 \text{ --- (1)}$$

$$f_y = 0$$

$$\therefore x - 2y = 0$$

putting the value of eqⁿ (1)

$$\therefore x - 2(3x^2) = 0$$

$$\Rightarrow x - 6x^2 = 0$$

$$\Rightarrow x(1 - 6x) = 0$$

$$\Rightarrow x = 0, 1/6$$

Again, (1) $\Rightarrow y = 3x^2$

putting the value of x

when $x = 0$, $y = 3(0)^2$
 $\therefore y = 0$

when $x = 1/6$, $y = 3(1/6)^2$
 $\therefore y = 1/12$

critical points: $(0, 0)$; $(1/6, 1/12)$

Critical point	$f_{xx}(x_0, y_0)$	$f_{yy}(x_0, y_0)$	$f_{xy}(x_0, y_0)$	$D = f_{xx}f_{yy} - f_{xy}^2$
$(0, 0)$	0	-2	1	-1
$(1/6, 1/12)$	-1	-2	1	1

At $(0, 0)$, $D < 0$, therefore there is saddle point at $(0, 0)$.

At $(1/6, 1/12)$, $D > 0$ and $f_{xx}(x_0, y_0) < 0$, therefore there is relative maximum at $(1/6, 1/12)$.

16. $f(x, y) = xy + \frac{2}{x} + \frac{4}{y}$
 $= xy + 2x^{-1} + 4y^{-1}$

$$f_x = y - 2x^{-2} + 0$$

$$= y - \frac{2}{x^2}$$

$$f_x = x + 0 - 4y^{-2}$$

$$= x - \frac{4}{y^2}$$

$$f_x = 0$$

$$\therefore y - \frac{2}{x^2} = 0$$

$$\Rightarrow y = \frac{2}{x^2} \text{ --- (1)}$$

$$f_y = 0$$

$$x - \frac{4}{y^2} = 0$$

$$\Rightarrow xy^2 - 4 = 0$$

putting the value
of eqn (1)

$$x\left(\frac{2}{x^2}\right)^2 - 4 = 0$$

$$\Rightarrow \frac{4}{x^3} - 4 = 0$$

$$\Rightarrow 4 - 4x^3 = 0$$

$$\Rightarrow x = 1$$

Again, (1) $\Rightarrow y = \frac{2}{x^2}$

putting the value of x

$$\therefore y = \frac{2}{1^2} = 2$$

critical point (1, 2)

$$f_{xx} = 0 + 4x^{-3} = \frac{4}{x^3}$$

$$f_{xy} = 1$$

$$f_{yy} = 0 + 8y^{-3} = \frac{8}{y^3}$$

Critical point (x_0, y_0)	$f_{xx}(x_0, y_0)$	$f_{yy}(x_0, y_0)$	$f_{xy}(x_0, y_0)$	$D = f_{xx}f_{yy} - f_{xy}^2$
(1, 2)	4	1	1	3

At (1, 2), since $D > 0$ and $f_{xx}(x_0, y_0) > 0$,
therefore there is a relative minimum
at (1, 2)

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$$f(x,y) = e^{-(x^2+y^2+4x)}$$

$$f_x = e^{-(x^2+y^2+4x)} x \cdot (+2x+4)$$
$$= -(2x+4) e^{-(x^2+y^2+4x)}$$

$$f_y = e^{-(x^2+y^2+4x)} x \cdot (-2y)$$
$$= -2y e^{-(x^2+y^2+4x)}$$

$$\therefore f(x) = 0 ; f(y) = 0$$

$$-(2x+4) e^{-(x^2+y^2+4x)} = 0$$

$$\therefore -(2x+4) = 0 \text{ or, } e^{-(x^2+y^2+4x)} = 0$$

$$\Rightarrow -2x-4=0 \text{ or, } N/A$$

$$\Rightarrow x = -2$$

Critical point $(-2,0)$

$$-2y e^{-(x^2+y^2+4x)} = 0$$

$$\therefore -2y = 0 \text{ or, } e^{-(x^2+y^2+4x)} = 0$$

$$\Rightarrow y = 0 \text{ or, } N/A$$

$$f_{xx} = -(2x+4) x e^{-(x^2+y^2+4x)} - (2x+4) + e^{-(x^2+y^2+4x)} x - 2$$

$$= \cancel{-(2+4)e^{-(1+0-4)}} x - \cancel{(-2+4)} + \cancel{e^{-(1+0-4)}} x - 2$$

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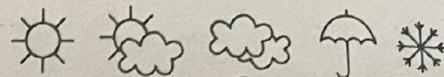
to the power
 $e = 0$

cannot be solved as

$\ln 0$ is Math Error.

Date

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$$\begin{aligned} \therefore f_{xx} &= -(-4+4) \times e^{-(4+0-\frac{8}{16})} \times -(-4+4) + e^{-(4+0-\frac{8}{16})} \times -2 \\ &= \cancel{0} - 109.2 \end{aligned}$$

$$\begin{aligned} f_{xy} &= -(2x+4) \times e^{-(x^2+y^2+4x)} \times -(2y) + e^{-(x^2+y^2+4x)} \times 0 \\ &= -\cancel{0}(-4+4) \dots \dots \dots \\ &= 0 \end{aligned}$$

$$\begin{aligned} f_{yy} &= -2y \times e^{-(x^2+y^2+4x)} \times (-2y) + e^{-(x^2+y^2+4x)} \times (-2) \\ &= 0 \times e^{-(4+0-\frac{8}{16})} \times 0 + e^{-(4+0-\frac{8}{16})} \times -2 \\ &= 0 + e^{-4} \times -2 \end{aligned}$$

$$= 0 + 54.6 \times -2$$

$$\begin{aligned} D &= \begin{pmatrix} -109.2 & 0 \\ 0 & -109.2 \end{pmatrix} \times (-109.2) - 0^2 \\ &= \cancel{0} 11,924 \end{aligned}$$

Since $D > 0$ and $f_{xx}(CP) < 0$, f has relative maximum